

Sugarcane

Saccharum officinarum — Pakistan's Major Sugar and Bioenergy Crop

A Complete Cultivation Guide for

Punjab · Sindh · Khyber Pakhtunkhwa · Balochistan

Compiled for Farmer Guidance & AI Advisory Use

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1. Introduction

Sugarcane is one of Pakistan's major cash crops, grown mainly in Punjab and Sindh's irrigated plains and supplying the country's large sugar milling industry along with providing raw material (gur/jaggery) for traditional sweeteners consumed widely across rural Pakistan. Unlike most other crops covered in this series, sugarcane is a long-duration crop, typically grown for ten to twelve months or more before harvest, and it can also be ratooned (regrown from the same root stock for one or more additional harvests) which affects its place in the cropping calendar. This guide explains how sugarcane is grown across Pakistan's provinces and how yield and sugar recovery can be improved.

2. Scientific Name

Sugarcane's scientific name is *Saccharum officinarum*, though most commercially grown modern varieties are complex hybrids sometimes classified under *Saccharum* spp. hybrids, belonging to the Poaceae (grass) family, the same family as wheat, rice, and maize.

3. Importance in Pakistan

Sugarcane is central to Pakistan's sugar industry, one of the largest agro-based industries in the country, supplying raw material to hundreds of sugar mills across Punjab and Sindh and supporting the livelihoods of millions of farming households. It is grown on roughly 1 to 1.2 million hectares and is also a key source of gur (traditional jaggery) consumed widely in rural areas, as well as being an increasingly important feedstock for ethanol production and bagasse-based cogeneration of electricity, adding to its economic significance beyond sugar alone. Because sugarcane is a long-duration crop with relatively high water requirements, its cultivation also has significant implications for Pakistan's water resource management, particularly in already water-stressed regions.

4. Major Producing Provinces and Districts

- Punjab: Faisalabad, Jhang, Toba Tek Singh, Sargodha, Multan, and Rahim Yar Khan — the largest sugarcane-growing region and home to the majority of the country's sugar mills.
- Sindh: Mirpurkhas, Tando Allahyar, Hyderabad, Thatta, and Badin — the second major sugarcane belt, with a well-established sugar milling industry of its own.
- Khyber Pakhtunkhwa: Mardan, Charsadda, and Nowshera — significant regional production supplying local sugar mills.
- Balochistan: Limited cultivation, mainly in areas with reliable irrigation such as Nasirabad division.

5. Climate Requirements

Sugarcane grows best in a warm, sunny climate with temperatures between 20°C and 35°C during active growth, and it requires a long frost-free growing period of at least ten months to reach full maturity and good sugar content. Adequate soil moisture throughout most of the growing season is essential, though a period of relatively drier, cooler weather before harvest helps promote sugar accumulation in the stalk, which is one reason winter harvest timing is important for sugar recovery in Pakistan's mills.

6. Soil Requirements

Sugarcane grows best in deep, well-drained, fertile loam to clay loam soils with good water-holding capacity, and a soil pH between 6.0 and 7.5, though it can tolerate somewhat alkaline soils common in parts of Punjab and Sindh. Because the crop remains in the field for many months and is often ratooned for additional years, good soil structure and fertility management over the long term are particularly important for sustaining yield across multiple harvests from the same field.

7. Land Preparation

Land preparation for sugarcane is more intensive than for most other crops given its long duration in the field: deep ploughing (often two to three times) followed by planking is used to create a fine, well-prepared seedbed, and furrows are made at appropriate spacing for planting setts (stem cuttings). Well-rotted farmyard manure, where available, is commonly incorporated ahead of planting to support the crop through its long growing period.

8. Recommended Varieties (Pakistan)

- CPF (Cane Punjab Faisalabad) series and other varieties developed by the Sugarcane Research Institute Faisalabad are widely recommended for Punjab's sugarcane belt, selected for good yield and sugar recovery.
- Varieties developed by the Sindh Sugarcane Research Institute Thatta and other Sindh-based research stations are recommended for that province's specific soil and climate conditions.
- Farmers should check the latest variety recommendations from provincial sugarcane research institutes each planting season, since new higher-yielding, disease-resistant, and better sugar-recovery varieties are periodically released to replace older ones that lose vigor or become more susceptible to disease over time.

9. Seed Rate

Sugarcane is propagated vegetatively using stem cuttings called setts rather than true seed. Recommended seed rate is about 25,000 to 30,000 three-budded setts per acre, requiring roughly 80,000 to 100,000 kg (80–100 tons) of seed cane per acre depending on cane thickness and the number of buds per sett used.

10. Seed Treatment

Setts should be treated by dipping in a fungicide solution (such as carbendazim or a similar recommended product) before planting to protect against sett rot and other soil-borne fungal diseases that can affect germination. Hot water treatment of setts (immersing in hot water at a controlled temperature for a set period) is also recommended in some programs to help control certain seed-borne diseases, particularly ratoon stunting disease.

11. Sowing Time (Province-wise)

- Punjab: Spring planting done February–March; autumn planting done September–October, which is often preferred where feasible since it generally gives a longer growing period and higher yield than spring planting.
- Sindh: Planted February–March for the spring crop; autumn planting in September–October is also practiced.
- Khyber Pakhtunkhwa: Planted February–March, aligned with the province's climate.
- Balochistan: Planted February–March in irrigated areas.

12. Sowing Methods

Setts are planted in furrows about 75–90 cm apart, laid end to end or slightly overlapping within the furrow, and then covered with soil to an appropriate depth, followed by irrigation to promote germination from the sett buds. Proper furrow spacing and sett placement are important for good early stand establishment, since gaps in the row directly reduce eventual yield and cannot always be fully compensated for by later tillering.

13. Fertilizer Recommendations (NPK)

A general recommendation for sugarcane is 150–200 kg Nitrogen, 90–100 kg Phosphorus (P₂O₅), and 60–75 kg Potash (K₂O) per acre, reflecting the crop's high nutrient demand given its long growing season and substantial biomass production. Full phosphorus and potash and about one-third to one-half of nitrogen are applied at or soon after planting, with the remaining nitrogen split into two or three top-dressings during the tillering and grand growth stages, since nitrogen availability during these stages strongly influences final cane yield.

14. Irrigation Schedule

Sugarcane requires regular irrigation throughout most of its long growing season, typically every 10–15 days depending on soil type and season, with the germination and early tillering stages and the grand growth (rapid stem elongation) stage being particularly sensitive to water stress. Irrigation is generally reduced or stopped a few weeks before harvest to help promote sugar accumulation in the stalk and ease harvesting operations.

15. Weed Management

Because sugarcane grows relatively slowly in its first few months before canopy closure, effective weed control during this early period is critical, typically achieved through two to three hoeings or inter-row cultivation combined with earthing up (mounding soil around the base of the plants), which also helps support the crop against lodging later in the season. Pre-emergence and post-emergence herbicides approved for sugarcane are increasingly used on larger commercial farms to reduce the substantial labor requirement of manual weeding over the crop's long growing period.

16. Insect Pests

Major insect pests of sugarcane in Pakistan include the sugarcane borer (stem borer, causing dead hearts in young cane and reduced sugar content in mature cane), pyrilla (a sap-sucking pest that can cause serious defoliation in outbreak years), and termites, which can damage both setts at planting and mature cane stalks. Integrated management includes resistant varieties, timely planting, removal and destruction of infested cane, conservation of natural predators (particularly important for pyrilla control), and need-based insecticide application during outbreak conditions.

17. Diseases

Red rot is the most economically damaging disease of sugarcane in Pakistan, a fungal disease that can spread through infected setts and cause severe stalk rot and juice quality deterioration, sometimes forcing the abandonment of susceptible varieties in affected areas. Smut and ratoon stunting disease are also significant concerns, particularly for ratoon crops. Management relies heavily on resistant varieties, use of disease-free certified seed cane, hot water or fungicide treatment of setts, and crop rotation or field sanitation to reduce disease carryover between plantings.

18. Deficiency Symptoms

Nitrogen deficiency shows as pale yellow-green older leaves and reduced tillering and stalk growth. Phosphorus deficiency causes stunted growth and poor root development, particularly affecting early establishment. Potassium deficiency appears as yellowing and drying along leaf margins and is particularly important in sugarcane since adequate potassium supports proper sugar translocation and accumulation in the stalk. Iron and manganese deficiencies can occur on the alkaline, calcareous soils common in parts of Punjab and Sindh, showing as interveinal yellowing of young leaves.

19. Harvesting

Sugarcane is ready for harvest when the crop reaches full maturity, generally indicated by reduced growth, drying of lower leaves, and peak sugar (sucrose) content in the stalk, typically 10 to 12 months after planting depending on variety and planting time, with harvest timing in Pakistan usually concentrated in the cooler winter months (November to March) when sugar mills are in operation. Harvesting is predominantly done by hand-cutting close to the ground level to maximize cane recovery, since the base of the stalk contains a significant portion of the accumulated sugar.

20. Storage

Unlike grain crops, sugarcane is highly perishable after cutting and should be transported to the sugar mill and crushed as quickly as possible, ideally within 24 to 48 hours of harvest, since delays lead to significant sugar (sucrose) losses due to inversion and drying of the cut stalks. This perishability makes efficient harvest scheduling and transport logistics between farms and mills a critical part of sugarcane's post-harvest handling in Pakistan, and delays caused by mill queues during peak crushing season are a recurring practical challenge for farmers.

21. Expected Yield in Pakistan

The national average sugarcane yield in Pakistan is around 55–65 tons per hectare, though this varies considerably by province and management level, with Punjab and Sindh's better-managed commercial farms often exceeding 80–90 tons per hectare using improved varieties, proper fertilization, and adequate irrigation. Sugar recovery rates (the percentage of sugar extracted per ton of cane crushed) also vary and are an important additional yield indicator alongside raw cane tonnage, since higher recovery directly benefits both farmers (through better pricing in some contract structures) and mills.

22. Government Recommendations

The Sugarcane Research Institutes in Faisalabad (Punjab) and Thatta (Sindh) lead variety development and agronomic research for the crop, working with provincial agriculture departments and the sugar milling industry to promote certified disease-free seed cane, balanced fertilization, and integrated pest and disease management. Government policy plays a particularly direct role in the sugarcane sector through the annual announcement of support prices for cane, which significantly affects farmer planting decisions and the broader dynamics between growers and sugar mills each season.

23. Modern Farming Practices

Improved seed cane multiplication programs to ensure availability of disease-free, high-yielding varieties, along with wider adoption of proper furrow irrigation and balanced fertilization guided by soil testing, are among the key modern practice improvements being promoted in Pakistan's sugarcane sector. Mechanical planting and harvesting are gradually being introduced on some larger farms, though hand harvesting remains dominant given labor availability and the crop's importance as a source of rural employment during the winter crushing season.

24. Precision Agriculture

Precision approaches being introduced in sugarcane include soil testing to guide fertilizer application over the crop's long growing season, drip irrigation trials aimed at improving water-use efficiency in this water-intensive crop, and satellite or drone-based monitoring being piloted by some research institutes and larger commercial farms to track crop growth and detect disease or nutrient stress earlier across large sugarcane blocks.

25. Sustainable Farming

Given sugarcane's high water requirement and long growing season, sustainability efforts focus on improving irrigation efficiency, including exploring drip irrigation and better furrow irrigation management to reduce water waste, along with proper ratoon management to extend productive cane life without excessive fresh planting, which reduces both cost and land preparation-related soil disturbance. Utilizing bagasse (the fibrous residue after juice extraction) for cogeneration of electricity at sugar mills, now increasingly common in Pakistan, also adds a valuable renewable energy dimension to the crop's overall sustainability profile.

26. Regional Notes

Punjab's extensive sugarcane belt around Faisalabad and Jhang, along with Sindh's cane-growing districts near Mirpurkhas, together anchor Pakistan's sugar milling industry, with the annual government-announced support price for cane playing a major role in shaping farmer decisions and grower-mill relations each season in both provinces. Continued attention to disease-free seed cane, balanced fertilization, and irrigation efficiency will remain central to sustaining and improving the productivity of this long-duration but economically vital crop across Pakistan's cane-growing regions.